Explainable AI for Heart Disease Prediction Using Public Healthcare Data

Kacper Kowalski
BSc (Hons) Data Science and Artificial Intelligence
Bournemouth University
2025/2026
Supervisor: Dr Emili Balaguer-Ballester
Word Count: 5,177 (Excluding Figures, Tables, References, and Appendices)

Abstract

Cardiovascular disease remains one of the leading causes of global mortality, making early detection and risk assessment important healthcare priorities. Machine learning methods have shown potential for heart disease prediction, but many high-performing models have limited transparency, reducing their suitability for decision support where explanations are needed. This project develops and evaluates an explainable machine learning system for heart disease prediction using the UCI Cleveland Heart Disease dataset.

Four classification models were implemented: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Model performance was evaluated using accuracy, precision, recall, F1-Score, ROC-AUC, and five-fold cross-validation. Explainability was evaluated using SHAP and LIME at both global and local levels, with Spearman‘s rank correlation used to measure agreement between feature importance rankings. An interactive Streamlit dashboard was also developed to present predictions, technical explanations and plain English summaries.

The results showed that Logistic Regression achieved the highest ROC-AUC, and XGBoost achieved the highest fixed-threshold classification performance. SHAP analysis found consistent high-ranking features across models, including chest pain type, thalassemia, number of major vessels, and ST depression, SHAP and LIME showed very high global agreement, with correlations above 0.93, but local agreement varied, especially with Logistic Regression. The Decision Tree was evaluated by itself using rule transparency and SHAP agreement. The project demonstrates that explainable models can support heart disease prediction, but limitations remain because of dataset size, lack of clinical validation, and the risk of overinterpreting post hoc explanations.

Acknowledgements

I would like to thank my supervisor, Dr Emili Balaguer-Ballester, for guidance and feedback throughout the project. I would also like to thank Bournemouth University and the Department of Computing for the academic support during the process.

Generative AI tools were used to help with spelling, grammar, structure, and clarity during the preparation of this report. All technical implementations, experiments, analysis, interpretation of results, and academic judgement were done by the author.

Generative AI was also used to support code review, debugging, and suggestions. All suggestions were reviewed, modified when needed, tested, and integrated by the author.

Introduction

1.1 Background & Motivation

Cardiovascular disease remains one of the leading causes of mortality globally, making early detection critical for improving patient outcomes and reducing healthcare burden. Machine learning models have increasingly been applied to support diagnosis by identifying patterns in clinical data and achieving strong predictive performance.

However, many high-performing models, especially ensemble and deep learning models, operate as opaque systems. This lack of transparency presents a barrier to the adoption of artificial intelligence in healthcare, where clinicians need transparent and justifiable decision-making.

The artefact produced by this project is an interactive dashboard that allows users to enter patient data, compare model predictions, and view SHAP, LIME, and plain English explanations.

1.2 Problem Statement

Although machine learning models have achieved strong performance in heart disease predictions, predictive accuracy is not enough for healthcare decision support. Clinicians and patients need explanations that are understandable, consistent, and aligned with the model's reasoning (Tonekaboni et al., 2019). Many existing heart disease prediction studies focus mainly on classification performance and provide limited evaluation of explanation methods and whether they are stable or meaningful outputs.

This project addresses that gap by developing a heart disease prediction artefact that combines machine learning models with explainability methods. The project evaluates not only predictive performance but also the consistency of SHAP explanations across model types, the agreement between SHAP and LIME at both global and local levels, and the feasibility of technical XAI being translated into plain English explanations for non-technical users.

1.3 Research Questions

1. RQ1: Do different machine learning models produce consistent feature importance rankings when analysed using SHAP?
2. RQ2: Do SHAP and LIME produce consistent explanations for the same model at both global and local levels?
3. RQ3: How can XAI outputs be translated into plain English explanations for non-technical users?

1.4 Aims and Objectives

Aim

The aim of this project is to develop and evaluate an explainable machine learning system for heart disease prediction using public healthcare data.

Objectives

- To review existing machine learning and explainable AI approaches for heart disease prediction.
- To preprocess and analyse a public heart disease dataset.
- To implement and compare Logistic Regression, Decision Tree, Random Forest, and XGBoost models.
- To evaluate model performance using accuracy, recall, F1-Score, ROC-AUC, and cross-validation.
- To apply SHAP and LIME to interpret model predictions.
- To develop a dashboard that presents predictions and explanations in an understandable format.

Literature Review

2.1 Machine Learning for Cardiovascular Disease Prediction

Cardiovascular disease (CVD) is the leading cause of death globally, accounting for an estimated 17.9 million deaths a year according to the World Health Organization (WHO, 2021). Early and accurate detection of heart disease is therefore a critical clinical priority. Machine learning has emerged as a promising approach to support early diagnosis, with numerous studies reporting strong predictive performance on publicly available datasets.

The UCI Cleveland Heart Disease Dataset, introduced by Detrano et al. (1989), has become a widely used benchmark for evaluating classification models in this domain. Gradient boosting methods such as XGBoost, have demonstrated especially strong performance because of their ability to capture non-linear feature interactions (Chen & Guestrin, 2016).

However, a consistent limitation across this body of work is the prioritisation of predictive accuracy instead of interpretability. Many studies report evaluation metrics such as accuracy, precision, recall and AUC without examining what the model had learned or whether its reasoning aligns with clinical knowledge. This gap between performance and transparency has been identified as a barrier to the adoption of machine learning in real environments (Obermeyer & Emanuel, 2016).

2.2 Explainable AI in Healthcare

Explainable Artificial Intelligence (XAI) refers to methods and techniques that make the outputs of machine learning models interpretable to humans. In healthcare, it is not only desirable but essential. Clinicians are required to justify their decisions to patients and colleagues, and a prediction system that cannot explain its reasoning is difficult to use in workflow (Holzinger et al., 2017).

XAI methods can be categorized as inherently interpretable or post-hoc. Inherently interpretable models such as logistic regression and decision tree have explanations as a natural product of their structure, a decision tree's branching logic can be read as a set of rules. Post-hoc methods like SHAP and LIME are applied after training to approximate the reasoning of more complex models. The trade-off between these is transparency vs predictive power. Prior work has also shown that explanations can influence user trust in clinical decision support systems, but poorly designed explanations can increase reliance or reduce confidence in correct outputs (Bussone et al., 2015; Cai et al., 2019).

2.3 SHAP (Shapley Additive Explanations)

SHAP was introduced by Lundberg and Lee (2017) as a unified framework for interpreting machine learning model predictions. The method is grounded in cooperative game theory, specifically the concept of Shapley values originally proposed by Shapley (1953). In this framework, each feature is calculated as its average marginal contribution across all possible subsets of features.

SHAP is useful in this project because it provides global feature importance and local explanations for individual predictions. It also allows explanations to be compared across models.

In practice, computing exact Shapley values is computationally intractable for large feature sets. So, Lundberg and Lee introduced TreeSHAP, an efficient algorithm for tree-based models such as Random Forest and XGBoost that computes exact SHAP values in polynomial time (Lundberg et al., 2020). For linear models, LinearSHAP gives an exact and efficient solution based on the model's coefficients and feature covariance structure.

2.4 LIME (Local Interpretable Model-Agnostic Explanations)

LIME was introduced by Ribeiro et al. (2016) as a model-agnostic post-hoc explanation method. Unlike SHAP which uses game theory, LIME approximates the behaviour of a complex model locally around a specific instance using a simpler interpretable model, usually a linear one.

For an individual instance, LIME samples nearby examples and fits a simple local model to approximate the original model's behaviour.

A key strength of LIME is its model-agnostic nature where it can be applied to any classifier that produces probability outputs regardless of its internal structure. This makes it especially useful in comparative studies where multiple model types are evaluated.

LIME has limitations. Because it relies on random sampling in each instance, repeated runs on the same instance can have different explanations known as instability (Alvarez-Melis and Jaakkola, 2018). This raises concerns about reliability in clinical settings where consistent explanations are important for trust. Also, LIME's local approximation can be a poor fit for regions of the feature space where the true decision boundary is non-linear.

2.5 SHAP vs LIME – Existing Comparisons

Several studies have compared SHAP and LIME in terms of explanation consistency, fidelity and usability. Bhatt et al. (2020) evaluated XAI methods in healthcare and found that SHAP produced more stable and grounded explanations than LIME, especially for tree models. Lundberg et al. (2020) argued that SHAP's consistency guarantees its suitability for clinical applications.

However, the relationship between SHAP and LIME is not straightforward. Studies have found that the two methods can produce meaningfully different feature importance rankings for the same model and instance, raising the question of which explanation a person should trust when the methods disagree (Slack et al., 2020). This is especially noticeable at the local level for individuals as the features aren't averaged as in global features.

Most importantly, existing comparisons have been conducted on benchmarks rather than the context of heart disease prediction. This project therefore contributes as an empirical evaluation of SHAP vs LIME specifically on the UCI Cleveland dataset, at both local and global levels, across four models of varying complexity.

Methodology

3.1 Dataset

This project uses the UCI Cleveland Heart Disease Dataset, one of the most widely used benchmarks in cardiovascular disease prediction research (Detrano et al., 1989). The dataset was developed at the Cleveland Clinic Foundation and contains 303 records with 13 features and a target variable indicating the presence or absence of heart disease.

The original target variable takes integer values from 0 to 4, where 0 is no heart disease and 1-4 are the degrees of severity. For this project, they have been collapsed into 0 and 1, no disease, and heart disease present.

Six records containing missing values such as "?" were removed, making the final dataset have 297 instances. The class distribution was 160 negative cases (53.9%) and 137 positive cases (46.1%) making the dataset a relatively balanced dataset and not needing resampling techniques.

The dataset contains 13 clinical features, including continuous variables such as age, cholesterol, resting blood pressure and maximum heart rate, and categorical variables such as chest pain type, thalassemia and exercise-induced angina. A full description of all features and encodings is provided in Appendix A.

3.2 Preprocessing

All preprocessing steps were implemented in a centralised setup to ensure consistency across all model training and evaluation scripts. Missing values were identified and replaced from “?” to NaN and then removed. All features were cast to float to make them compatible with scikit-learn estimators.

The dataset was split between training and test sets using an 80/20 ratio with stratification on the target variable to preserve class proportions in both sets, this resulted in 237 training instances and 60 test ones. A fixed seed of 42 was used to have full reproducibility.

Feature scaling using StandardScaler was applied to the Logistic Regression model because feature magnitudes affect coefficient estimation. Tree-based models such as Decision Tree, Random Forest, and XGBoost are invariant to feature scaling and were trained on unscaled data.

Table 3.1 summarises the preprocessing steps used before model training. These steps were consistent across models to support fair comparison.

Step	**Action Taken**	**Justification**
Missing values	Replaced ? with missing values and removed affected rows	Required because six records contained incomplete feature values
Target conversion	Converted target values 1-4 into 1	Creates binary classification task: disease vs no disease
Data type conversion	Converted all features to float	Ensures compatibility with scikit-learn and XGBoost
Train/test split	80/20 stratified split	Preserves class distribution across train and test sets
Random seed	random_state = 42	Ensures reproducibility
Scaling	StandardScaler used for Logistic Regression only	Linear models are scale-sensitive; tree models are not

Using the same cleaned dataset and split ensured that performance differences were caused by the model instead of inconsistent preprocessing.

3.3 Model Selection and Justification

Four classification models were selected to represent different levels of interpretability and predictive complexity, letting the project examine the trade-offs.

Logistic Regression was chosen as the baseline linear model. It provides direct insight into the direction and magnitude of each feature’s effect on the prediction, making it interpretable by design.

A Decision Tree was selected as the fully transparent baseline. The decision tree model requires no post-hoc explanation as its logic can be read directly as a set of human interpretable clinical rules. This makes it valuable as a ground truth reference for evaluating the faithfulness of SHAP explanations.

Random Forest was selected as a high-performance ensemble method. By aggregating predictions across a large number of decision trees, Random Forest has strong generalisation but sacrifices interpretability.

XGBoost was selected as a gradient boosting method which is known for its performance on tabular data (Chen & Guestrin, 2016). Like Random Forest, its ensemble structure makes it opaque.

All tree-based models use class_weight=”balanced” or scale_pos_weight to account for the slight class imbalance in the dataset, with particular motivation from the healthcare context where false negatives have a higher cost than false positives. Hyperparameters for Decision Tree, Random Forest, and XGBoost were tuned using fivefold cross validation GridSearchCV with AUC as the scoring method. Logistic regression was used as a fixed baseline model with max_iter=1000 and class_weight=”balanced”.

3.4 Evaluation Metrics

Model performance was evaluated using five metrics: accuracy, precision, recall, F1 score and ROC curve (AUC). While accuracy is a useful measure of performance, it can be misleading in class imbalance. AUC was used as the primary evaluation metric, as it measures the model's ability to discriminate between classes across all classification thresholds and is robust to class imbalance.

Recall is treated as a secondary metric because of clinical context of this project. In heart disease prediction, a false negative carries a high cost as a missed diagnosis can delay treatment and have worse patient outcomes. All models were optimised to maintain high recall without sacrificing overall performance.

Fivefold cross validation was used in all models on the training set to provide a more useful estimate of generalisation performance than a single train test split which is important especially for a dataset with a relatively small size.

Because the held-out contained only 60 instances, test-set metrics were interpreted cautiously. Cross-validation was used alongside the test set to give a better view of model performance.

3.5 XAI Approach

SHAP explanations were generated using the TreeExplainer for all tree-based models and the LinearExplainer for Logistic Regression. Both explainers compute exact SHAP values rather than approximations. For multi output tree models that return SHAP values for both classes, the class 1 (disease) SHAP values were extracted across all analyses.

LIME explanations were generated using the LimeTabularExplainer with discretize_continuous=True, which converts continuous features into human readable conditional ranges such as age>55 instead of raw values. This was chosen to try and improve the accessibility of explanations for non-technical users.

To compare SHAP and LIME agreement both globally and locally, mean absolute SHAP values were computed across the test set and compared against mean absolute LIME feature importances averaged across 50 test instances. For local comparison, both methods were applied to the same representative patient, a true positive case correctly identified as having heart disease. Spearman rank correlation was used as the agreement metric in all comparisons, as it captures monotonic relationships between feature importance rankings without assuming a linear relationship.

The LIME stability analysis, which was run five times on the same instance showed low variance in feature weight ranking, with a mean standard deviation of 0.0026 across features. This suggests that instability was not a significant concern for this dataset.

System Design and Implementation

4.1 System Architecture

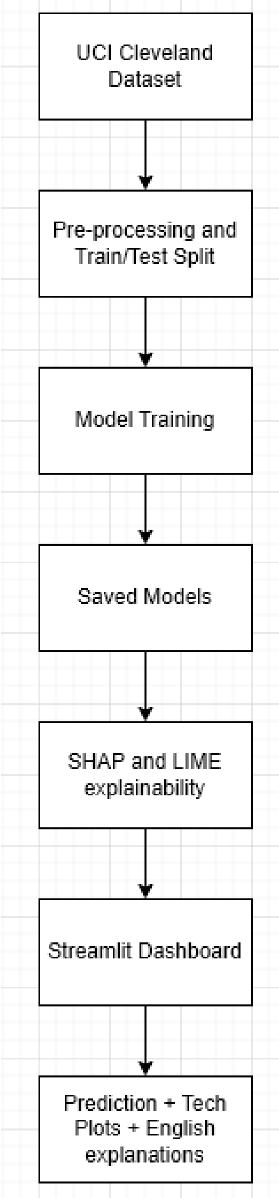
The system developed in this project has three integrated components: a data preprocessing and model training pipeline, an explainability module, and an interactive web-based dashboard.

The preprocessing pipeline is centralised in a single setup module that loads and cleans the raw dataset, performs the train test split, fits the feature scaler, and exports all shared objects for use by downstream scripts. This design makes sure that all models are trained and evaluated on identical data splits and that the scaler fitted on training data is reused consistently, preventing data leakage.

The explainability module has three analysis scripts. The first applies SHAP to all four models and generates global and local explanation outputs. The second performs a cross model SHAP comparison computing mean absolute SHAP values for each model and evaluating feature importance agreement using Spearman rank correlation. The third performs a direct SHAP vs LIME comparison at both global and local levels, producing agreement scores and visualisations for the results analysis.

The dashboard loads all models and the fitted scaler at startup, accepts patient data as input through a sidebar, generates predictions and explanations in real time, and presents results in both technical and plain English formats.

Figure 4.1 shows the overall architecture of the artefact, from dataset preprocessing to prediction and explanation output.



4.2 Model Implementation

All four models were implemented using scikit-learn's unified estimator API, with the exception of XGBoost which uses its Python library. This simplifies evaluation and ensures all models expose the same predict and predict_proba interfaces needed by the LIME explainer.

Model	Role in Project	Reason for Selection	Interpretability Level
Logistic Regression	Baseline linear model	Provides strong interpretability through coefficients	High

Decision Tree	Transparent rule-based model	Produces readable decision paths and supports interpretability validation	High
Random Forest	Ensemble comparison model	Strong tabular performance but less transparent	Medium / Low
XGBoost	Gradient boosting model	Often performs strongly on structured datasets	Low

This selection allowed the project to compare interpretable models with more opaque ensemble methods that benefit from post-hoc explanations.

Table 4.2 summarises the main hyperparameter search spaces and selected parameters.

Model	**Parameters Searched**	**Selected Parameters**
Logistic Regression	max_iter, class_weight	max_iter=1000, class_weight="balanced"
Decision Tree	max_depth, min_samples_split, criterion	max_depth=3, min_samples_split=2, criterion="gini"
Random Forest	n_estimators=[100,200,300], max_depth=[3,5,10,None], min_samples_split=[2,5,10]	max_depth: 5, min_samples_split: 5, n_estimators: 300
XGBoost	n_estimators=[100,200,300], max_depth=[3,4,5], learning_rate=[0.01,0.05,0.1]	learning_rate: 0.01, max_depth: 4, n_estimators: 300

AUC was used during tuning because it evaluates discrimination across classification thresholds, which is needed for a healthcare prediction task.

All models were trained with class imbalance handling. Logistic Regression and Decision Tree used scikit-learn's class weight=”balanced” parameter which adjusts sample weights inversely proportional to class frequencies. Random Forest used the same approach, XGBoost used the scale_pos weight parameter calculated as the ratio of negative to positive training instances (1.174) which achieves an equivalent effect in XGBoost. This consistent treatment of class balance in all models ensures that comparisons of recall and F1 score are not changed by different sensitivity to the class distribution.

4.3 XAI Integration

The explainability module was designed with two objectives: to generate explanations of individual model predictions and to allow comparison of explanation methods across models. SHAP was used on all four models after training. LIME was applied to all apart from Decision Tree which was evaluated with its transparent decision path and a comparison between Gini importance and SHAP importance.

For SHAP, the choice of explainer was matched to the model type to ensure exact Shapley value comparison. TreeExplainer was used for Decision Tree, Random Forest and XGBoost models, exploiting the tree structure to compute exact SHAP values using the TreeSHAP algorithm (Lundberg et al., 2020). LinearExplainer was used for Logistic Regression, computing exact SHAP values based on the model's coefficients and the training data covariance structure.

A key design choice was the extraction of class 1 SHAP values for all multi output models. Tree based classifiers in scikit learn returns SHAP values for both classes at the same time. Producing a three-dimensional output array of shape (n_samples, n_features, n_classes). Consistently extracting the disease class SHAP values makes sure that all feature importance values reflect contribution towards the prediction, making results comparable across all models and interpretable in a clinical context.

For LIME, a single LimeTabularExplainer instance was configured with the training data as the background distribution. The discretize_continuous parameter was set to True in the dashboard and local explanation scripts, making continuous feature values into conditional range statements for readability. In the cross-method comparison scripts, it was set to False to get raw numerical importance values for the Spearman rank correlation analysis.

The SHAP versus LIME comparison was done at two levels. At the global level, mean absolute values were averaged across all 60 test instances, and the mean absolute LIME importances were averaged across 50 test instances sampled from the test set. At the local level, both methods were used to the same true positive patient to examine if they produced consistent explanations for an individual prediction. Spearman rank correlation was computed between the feature importance rankings produced by each method at both levels for all models.

4.4 Dashboard Design

The dashboard was implemented in Streamlit because it supports rapid development of interactive machine learning interfaces directly from Python.

The dashboard is structured around two primary user interactions. First the user enters the patient data in the sidebar with sliders and dropdown menus for all features. All input controls use human readable labels and value descriptions such as thalassemia using options such as "Normal" and "Fixed Defect" instead of codes such as 3, 6, or 7. This was done to target non-technical users who would be unfamiliar with encoding language.

Second, the user selects one or more models to run and clicks the Run Predictions button. The dashboard then generates predictions and explanations for each selected model and presents them in a structured layout. For each model, the dashboard displays three outputs: a plain English explanation made from SHAP values, a SHAP waterfall plot showing the contribution of each feature to the prediction, and a LIME bar chart showing the importances assigned by the local linear approximation. For the Decision Tree, the dashboard also adds the full decision path, the sequence of feature thresholds the model evaluated to reach the prediction also in readable step by step.

The plain English explanations system was designed as the main output for users. Rather than showing raw SHAP values or scores, the system translates each features contribution into a sentence. Features are described as “strongly”, “slightly” or “marginally” based on their SHAP value relative to the maximum SHAP value for that prediction. Human readable value descriptions are applied to all categorical features for example a chest pain value of 4 is described as “Asymptomatic” instead of 4. A risk summary sentence is generated based on the class and probability, and all explanations include a disclaimer saying that the system supports judgement instead of replacing it.

Figure 4.2 shows the Streamlit dashboard developed as the project artefact

The dashboard lets users enter patient values, select models, and view predictions in SHAP, LIME, and English explanations.

Results and Analysis

5.1 Model Performance

The performance of all four models was evaluated on the held-out test set using accuracy, precision, recall, F1-Score, and ROC-AUC with five-fold cross validation to assess general performance.

Table 5.1 shows the test performance of each model.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Interpretation
Logistic Regression	0.8333	0.8462	0.7857	0.8148	0.9498	Best overall ranking ability
Decision Tree	0.8000	0.7857	0.7857	0.7857	0.8220	Most interpretable but weakest performance
Random Forest	0.8333	0.8462	0.7857	0.8148	0.9464	Strong balance of performance and stability

XGBoost	0.8500	0.8800	0.7857	0.8302	0.9263	Best fixed-threshold classification performance

The results show that Logistic Regression achieved the highest ROC-AUC, while XGBoost achieved the best fixed threshold classification result. These results are similar with recent explainable heart disease prediction work on the Cleveland dataset. For example, Adekoya et al. (2025) evaluated an interpretable ensemble framework using SHAP and LIME and reported AUC of 0.899 and accuracy of 88.5% on the Cleveland dataset. However, direct comparison should be treated cautiously as validation strategies, preprocessing, and model design are different.

Table 5.2 shows five-fold cross-validation AUC on the training set

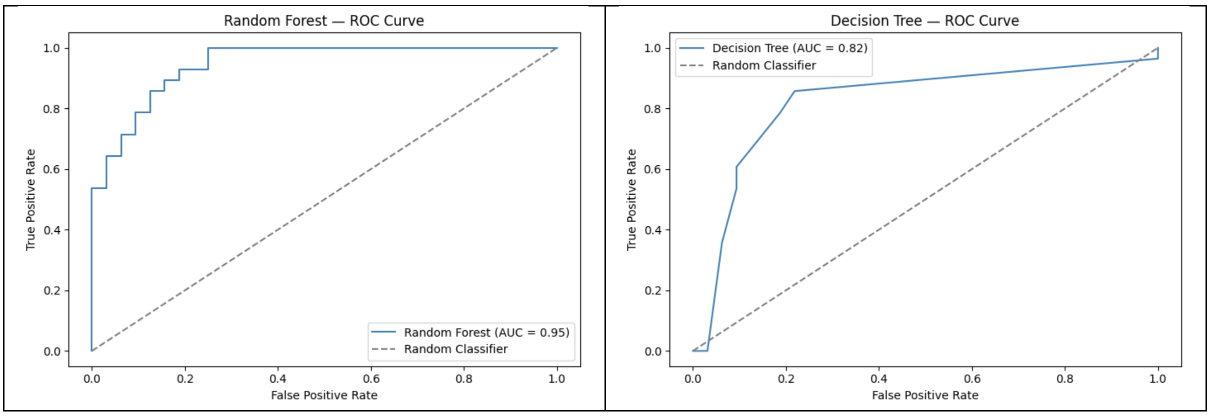
Model	**5-Fold CV AUC Mean**	**Standard Deviation**	**Comment**
Logistic Regression	0.8734	0.0901	Higher variance than Random Forest
Decision Tree	0.8374	0.0783	Lower but interpretable baseline
Random Forest	0.9020	0.0517	Strongest cross-validation stability
XGBoost	0.8869	0.0683	Strong performance but may be limited by dataset size

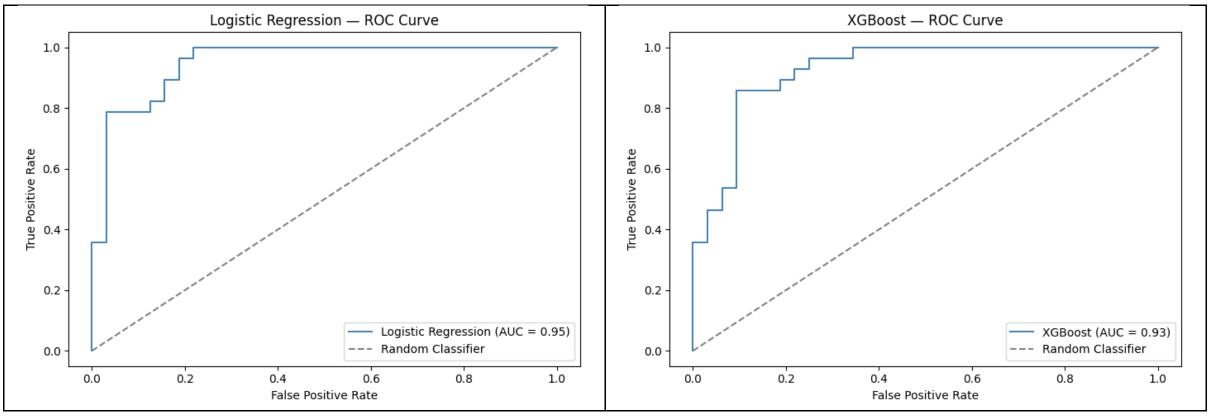
Random Forest achieved the strongest cross validation stability, suggesting stronger generalisation across folds.

XGBoost achieved the strongest fixed threshold performance, with the highest accuracy, precision, and F1-score. Logistic Regression achieved the highest ROC-AUC, suggesting that it ranked patients well across thresholds even if it has a simpler linear structure, random Forest showed the best cross-validation stability. The Decision Tree performed weakest overall but remains useful as a transparent baseline.

Models such as logistic regression have performance similar to opaque ensemble methods, suggesting that explainability can be prioritised without loss in predictive power.

Figure 5.1 shows the ROC curve performance across the four models.





5.2 SHAP Analysis

SHAP was used to analyse feature importance across all models allowing both global and cross model comparisons. The results show a high degree of consistency in feature importance rankings across different model types.

Table 5.3 Compares SHAP feature importance ranking across Logistic Regression, Random Forest, and XGBoost.

Feature	Logistic Regression Rank	Random Forest Rank	XGBoost Rank	Average Rank
cp	4	1	1	2.00
thal	2	2	2	2.00
ca	1	3	3	2.33
oldpeak	5	4	4	4.33
slope	6	5	7	6.00
sex	3	9	8	6.67
exang	7	7	11	8.33
age	13	8	5	8.67
thalach	11	6	9	8.67
chol	12	10	6	9.33
trestbps	8	12	10	10.00
restecg	9	11	12	10.67
fbs	10	13	13	12.00

The most consistently important features were chest pain type, thalassemia, number of major vessels, and ST depression.

Table 5.4 reports Spearman rank correlations between SHAP feature rankings.

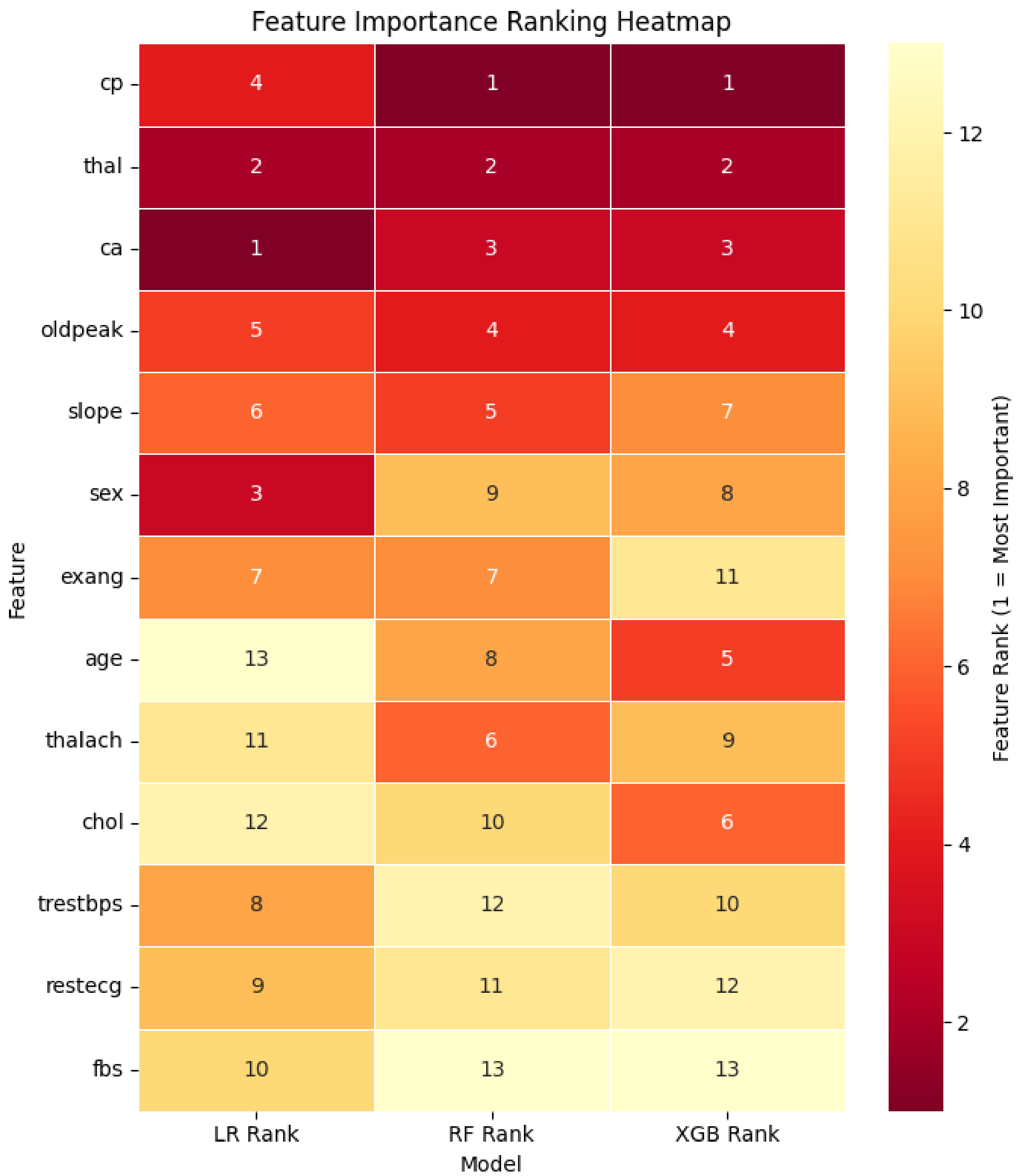
Model Pair	Spearman r	p-value	Interpretation
Logistic Regression vs Random Forest	0.6319	0.0205	Moderate significant agreement
Logistic Regression vs XGBoost	0.5000	0.0819	Moderate but not statistically significant

Random Forest vs XGBoost	0.8352	0.0004	Strong significant agreement
Average agreement	0.6557	N/A	Moderate to strong overall consistency

The most consistent important features were chest pain type, thalassemia, number of major vessels, and ST depression. The strongest agreement was between Random Forest and XGBoost, suggesting that tree-based models learned similar importance structures. Agreement involving Logistic Regression was weaker, suggesting that linear and nonlinear models represented data differently.

The high agreement between Decision Tree Gini importance and SHAP importance ($r = 0.9929$, $p < 0.001$) provides additional evidence that SHAP reflected the behaviour of the interpretable Decision Tree model.

Figure 5.2 visualises the feature ranking patterns across models



The heatmap shows that highly ranked features were more stable across models than lower ranked ones.

5.3 SHAP vs LIME Comparison

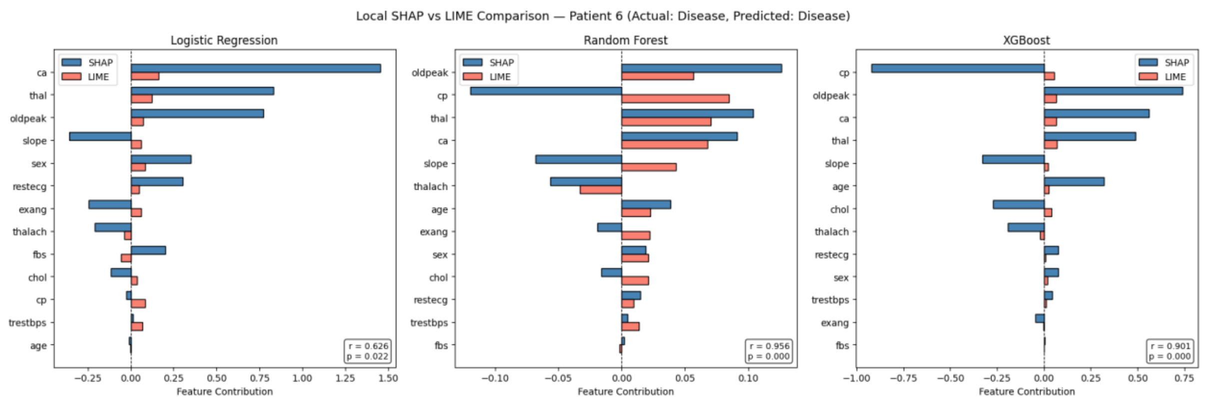
The agreement between SHAP and LIME was evaluated both locally and globally using Spearman rank correlation.

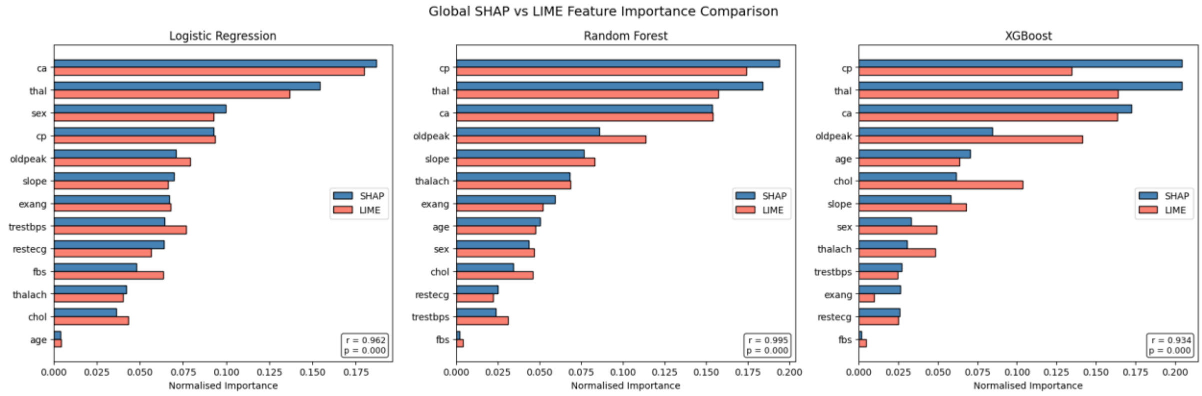
Table 5.5 compares SHAP and LIME feature ranking agreement at global and local levels.

Model	**Global Spearman r**	**Global p-value**	**Local Spearman r**	**Local p-value**	**Interpretation**
Logistic Regression	0.9615	0.0000	0.6264	0.0220	Strong global agreement, weaker local agreement
Random Forest	0.9945	0.0000	0.9560	0.0000	Very strong agreement at both levels
XGBoost	0.9341	0.0000	0.9011	0.0000	Very strong agreement at both levels

SHAP and LIME showed very high global agreement across all three models compared, with statistically important correlations above 0.93. Local agreement was more variable. Logistic Regression dropped to moderate local agreement, while Random Forest and XGBoost stayed high. This shows that explanation methods can appear consistent globally but still are different for individual patients, which is important in healthcare where decisions are made for individual patients.

Figure 5.3 visualises the difference between global and local SHAP-LIME agreement.





This supports the finding that explanation methods are more consistent when averaged across a dataset than when applied to individual predictions.

5.4 Plain English Explanations

The plain English system translates SHAP values into natural language statements such as “strongly increases risk”. It maps coded values into readable descriptions, for example describing chest pain value 4 as “Asymptomatic” and summarises if each factor raises or lowers predicted risk.

This improves accessibility for non-technical users, but it also introduces limitations. The threshold-based language system creates boundaries that might not reflect meaningful differences. A SHAP value just above or below a threshold can receive a different description, which can mislead users. Future work should validate if these descriptions are understandable and clinically appropriate.

Figure 5.4 shows an example of the plain English explanation generated.

5.5 Research Question Summary

Table 5.6 summarises how the results answer the project's research questions.

Research Question	Evidence Used	Finding
Do different models produce consistent SHAP feature rankings?	SHAP rankings and Spearman correlations	Yes, especially between Random Forest and XGBoost; top features were stable across models
Do SHAP and LIME produce consistent explanations?	Global and local Spearman correlations	Yes globally, but local agreement varied, especially for Logistic Regression
Can XAI outputs be translated into plain English?	Dashboard explanation system	Yes, SHAP values were converted into readable risk statements, though user testing is needed

Overall, the results show that the artefact met the project aim by combining predictive modelling with explainability evaluation and user explanation. RQ1 was addressed through cross-model SHAP ranking comparisons, which showed that key features were stable across model types. RQ2 was answered through SHAP and LIME agreement analysis, which showed strong global agreement but weaker local agreement for Logistic Regression. RQ3 was answered through the dashboards plain English system, although user testing would be needed to confirm how effectively users understand them.

Evaluation

6.1 Technical Evaluation

The technical evaluation shows that the artefact met its main aim by combining predictive performance with interpretable output. Logistic Regression, and XGBoost achieved similar performance, suggesting that model complexity did not improve quality on the dataset.

The explainability evaluation showed that global explanations were more stable than local explanations. SHAP looks appeared suitable as the main method because it created consistent global rankings and aligned well with the interpretable Decision Tree. LIME stayed useful for local explanations, but its outputs should be treated cautiously.

6.2 Artefact and Usability Evaluation

The qualitative evaluation focused on the usability and interpretability of the dashboard rather than validity. The system was assessed using three criteria: clarity of input, clarity of prediction output, and clarity of explanation output.

The plain English explanation system is the main qualitative contribution of the artefact. Instead of requiring users to interpret SHAP plots, the system converts the feature contributions into natural language statements describing whether a feature raises or lowers risk. This addresses the research question of whether XAI outputs can be translated into meaningful explanations for non-technical users.

However, the qualitative evaluation also found limitations. The explanations have not been evaluated with clinicians or patients, so their real-world interpretability cannot be confirmed. Also, plain English explanations can create a sense of certainty if users don't understand that SHAP values explain model behaviour instead of medical causality. For this reason, the dashboard has a disclaimer that the system should be used as support instead of replacing judgement.

6.3 Limitations

The main limitation of this project is the size and age of the dataset. The UCI Cleveland dataset only has 303 original records, reduced to 297 after removing missing values. Even though it is widely used as the benchmark, it is not representative of modern clinical populations and does not include the breadth of the demographic, lifestyle, imaging, medication, or variables used in real cardiovascular assessment.

A second limitation is that the system has not been clinically validated. The dashboard should be seen as a research artefact and demonstration tool, not as a medical system. The reported performance shows predictive capability on a benchmark dataset, but not safety, reliability or generalisability in a real environment.

The project is also limited by the use of post-hoc explanation methods. SHAP and LIME explain model behaviour, not true medical causality. A feature increasing predicted risk does not mean that the feature causes heart disease. This is important as users may overtrust explanations.

Another limitation is the treatment of categorical variables. Features such as chest pain type, thalassemia, resting ECG, and ST slope were kept in their original numeric encodings. This added consistency with the UCI dataset and supported comparison with common benchmarks, but it may cause models, such as Logistic Regression to treat categories as ordered numerical values. Future work should compare this approach with one-hot encoding or model pipelines that explicitly handle categorical features.

Finally, the qualitative evaluation did not include formal user testing with clinicians, or healthcare professionals, and even if the plain English system was designed to improve accessibility, its effectiveness would need to be evaluated through user studies.

Conclusion and Future Work

7.1 Summary of Contributions

This project developed an explainable machine learning artefact for heart disease prediction using public healthcare data. The artefact combines four classification models, SHAP and

LIME explainability methods, and an interactive dashboard that presents both technical and plain English explanations.

The main contribution is the evaluation of explanation consistency across models and methods, instead of prediction alone.

7.2 Key Findings

The results show that strong predictive performance can be achieved without relying on opaque models. Logistic regression achieved the highest ROC-AUC, while XGBoost achieved the strongest fixed threshold classification performance. This suggests that, for the Cleveland dataset, simpler and more interpretable models remain competitive with more complex ensemble methods.

The SHAP analysis showed that several features were consistently important across models, especially chest pain type, thalassemia, number of major vessels, and ST depression. This consistency supports the reliability of the learned patterns and suggests that the models were using clinically plausible predictors.

The SHAP and LIME comparison showed very high agreement at the global level, but weaker and more variable agreement at the local level. This is important because clinical decisions are made for individuals rather than average cases. The result supports the use of XAI as interpretability aids but also shows problems with local explanations.

7.3 Future Work

Future work should evaluate the system on larger and more recent healthcare datasets to test whether the findings generalise beyond the UCI Cleveland benchmark. External validation on a dataset with a different population would be important before it is considered clinically viable.

The dashboard should also be evaluated with real users, such as clinicians, students, and non-technical users. A formal usability study could assess whether the plain English explanations improve understanding, if users interpret the outputs correctly, and if the explanations increase or decrease trust.

Additional future work could include calibration analysis, threshold optimisation based on the cost of false negatives, and fairness analysis across groups. This would strengthen the healthcare relevance of the system and provide a more complete evaluation of model safety and reliability.

7.4 Personal Reflection

This project developed my understanding of the difference between building a model that performs well and building a system that can be meaningfully interpreted. Early in the project, accuracy was the main measure of success. However, the explainability analysis showed that strong performance is only one part of the problem, especially in healthcare.

Implementing SHAP and LIME across models also taught me that explanation methods should be treated critically rather than accepted automatically, as they can disagree on local levels.

References

Alvarez-Melis, D., & Jaakkola, T. S. (2018). On the robustness of interpretability methods. *arXiv*. https://arxiv.org/abs/1806.08049

Bhatt, U., Xiang, A., Sharma, S., Weller, A., Taly, A., Jia, Y., Ghosh, J., Puri, R., Moura, J. M. F., & Eckersley, P. (2020). Explainable machine learning in deployment. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 648–657. https://doi.org/10.1145/3351095.3375624

Bussone, A., Stumpf, S., & O'Sullivan, D. (2015). The role of explanations on trust and reliance in clinical decision support systems. *2015 International Conference on Healthcare Informatics*, 160–169. https://doi.org/10.1109/ICHI.2015.26

Cai, C. J., Jongejan, J., & Holbrook, J. (2019). The effects of example-based explanations in a machine learning interface. *Proceedings of the 24th International Conference on Intelligent User Interfaces*, 258–262. https://doi.org/10.1145/3301275.3302289

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.-J., Sandhu, S., Guppy, K., Lee, S., & Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *The American Journal of Cardiology, 64*(5), 304–310. https://doi.org/10.1016/0002-9149(89)90524-9

Holzinger, A., Biemann, C., Pattichis, C. S., & Kell, D. B. (2017). What do we need to build explainable AI systems for the medical domain? *arXiv*. https://arxiv.org/abs/1712.09923

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence, 2*(1), 56–67. https://doi.org/10.1038/s42256-019-0138-9

Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future: Big data, machine learning, and clinical medicine. *The New England Journal of Medicine, 375*(13), 1216–1219. https://doi.org/10.1056/NEJMp1606181

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should I trust you?”: Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144. https://doi.org/10.1145/2939672.2939778

Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 180–186. https://doi.org/10.1145/3375627.3375830

Tonekaboni, S., Joshi, S., McCradden, M. D., & Goldenberg, A. (2019). What clinicians want: Contextualizing explainable machine learning for clinical end use. *Proceedings of Machine Learning Research, 106*, 359–380.

World Health Organization. (2021). *Cardiovascular diseases (CVDs)*. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

Adekoya, A. A., Saeed, F. S., Ghaban, W. G., & Qasem, S. N. (2025). Ensemble learning approach with explainable AI for improved heart disease prediction. Frontiers in Pharmacology, 16, 1654681. https://doi.org/10.3389/fphar.2025.1654681

Appendices

Appendix A: Dataset Feature Descriptions

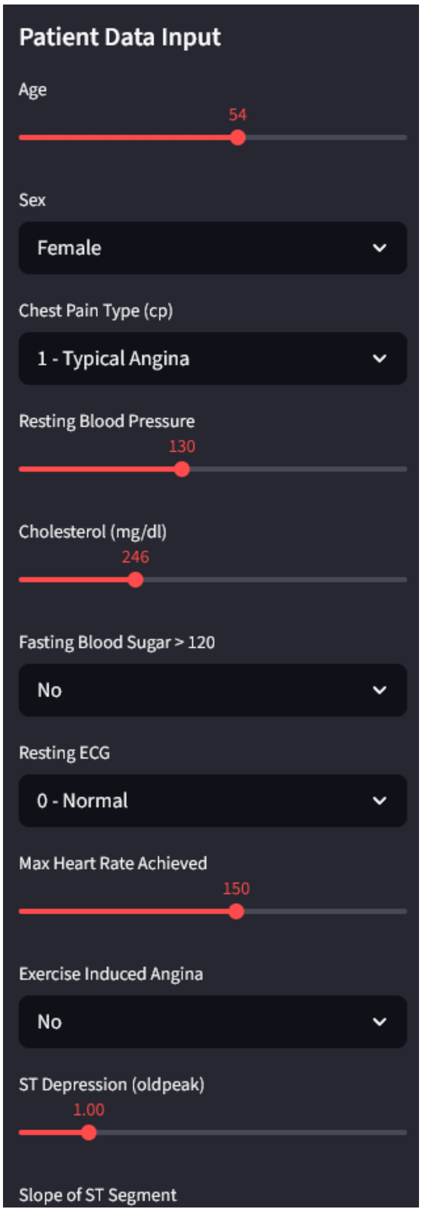
Feature	Description	Type	Example Values / Encoding
age	Age of patient	Continuous	Years
sex	Biological sex	Binary categorical	0 = female, 1 = male
cp	Chest pain type	Categorical	1 = typical angina, 2 = atypical angina, 3 = non-anginal pain, 4 = asymptomatic
trestbps	Resting blood pressure	Continuous	mm Hg
chol	Serum cholesterol	Continuous	mg/dl
fbs	Fasting blood sugar > 120 mg/dl	Binary categorical	0 = false, 1 = true
restecg	Resting ECG result	Categorical	0, 1, 2
thalach	Maximum heart rate achieved	Continuous	Beats per minute
exang	Exercise-induced angina	Binary categorical	0 = no, 1 = yes
oldpeak	ST depression induced by exercise	Continuous	Numeric
slope	Slope of peak exercise ST segment	Categorical	1, 2, 3
ca	Major vessels coloured by fluoroscopy	Ordinal / numeric	0-3
thal	Thalassemia result	Categorical	3 = normal, 6 = fixed defect, 7 = reversible defect
target	Heart disease diagnosis	Binary target	0 = no disease, 1 = disease present

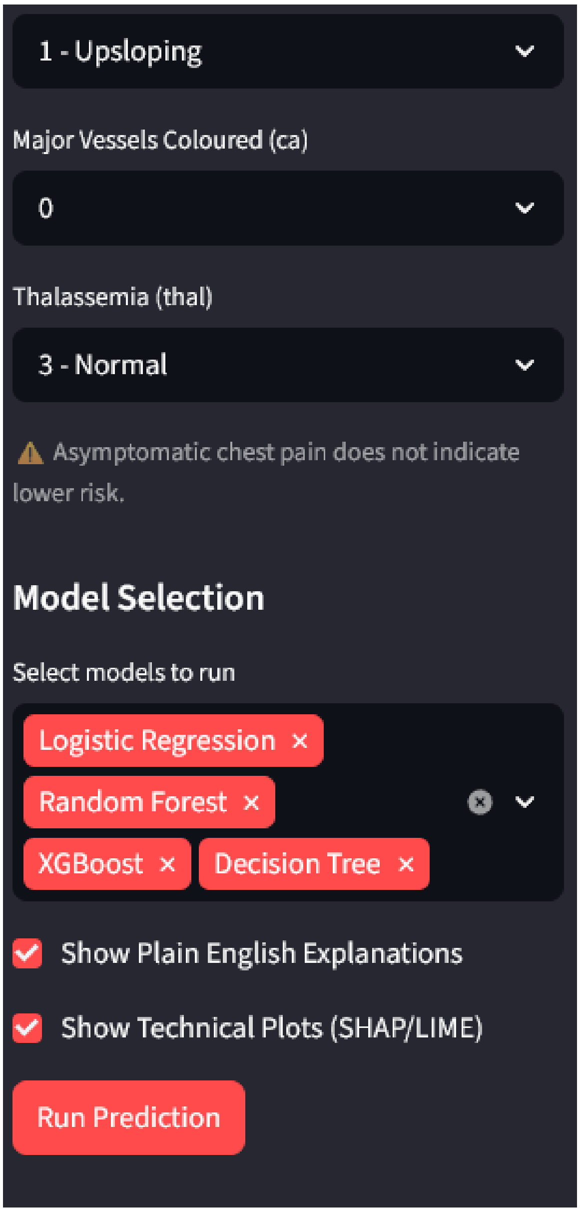
Appendix B: Hyperparameter Search Grid

Model	**Hyperparameter Search Grid**
Logistic Regression	Fixed Parameters: max_iter=1000, class_weight="balanced"
Decision Tree	max_depth=[3,4,5,6,None], min_samples_split=[2,5,10], criterion=["gini","entropy"]
Random Forest	n_estimators=[100,200,300], max_depth=[3,5,10,None], min_samples_split=[2,5,10]
XGBoost	n_estimators=[100,200,300], max_depth=[3,4,5], learning_rate=[0.01,0.05,0.1]

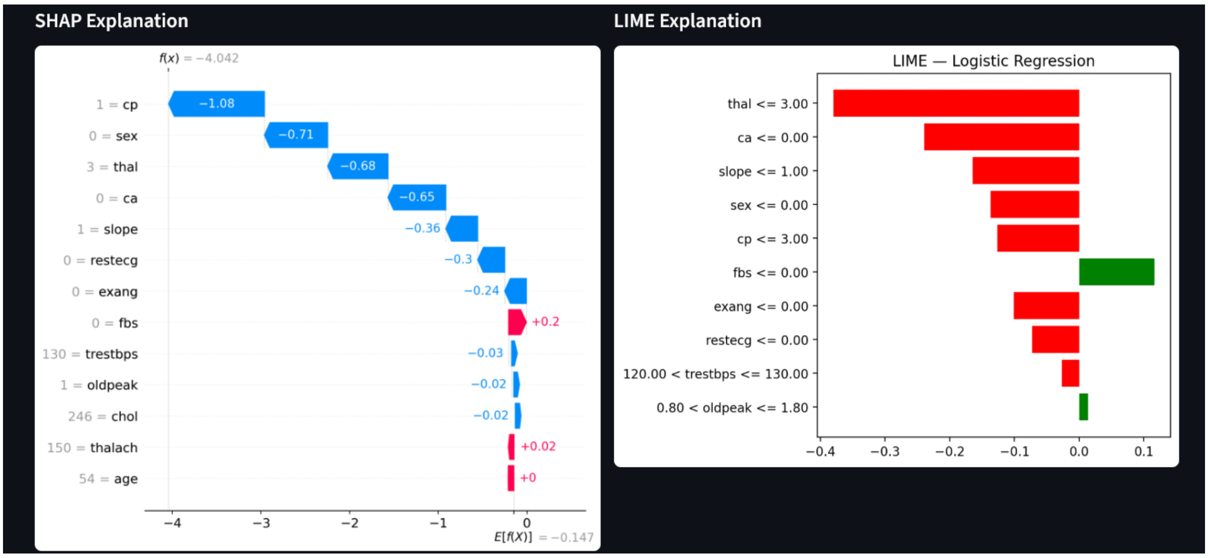
Appendix C: Additional Artefact Screenshots

Appendix C contains additional screenshots of the dashboard, including the patient input sidebar, model comparison, SHAP waterfall plots, LIME explanation, and plain English Explanations.


Patient Data Input
Age
54
Sex
Female
Chest Pain Type (cp)
1 - Typical Angina
Resting Blood Pressure
130
Cholesterol (mg/dl)
246
Fasting Blood Sugar > 120
No
Resting ECG
0 - Normal
Max Heart Rate Achieved
150
Exercise Induced Angina
No
ST Depression (oldpeak)
1.00
Slope of ST Segment



1 - Upsloping
Major Vessels Coloured (ca)
0
Thalassemia (thal)
3 - Normal
Asymptomatic chest pain does not indicate lower risk.
Model Selection
Select models to run
Logistic Regression
Random Forest
XGBoost
Decision Tree
Show Plain English Explanations
Show Technical Plots (SHAP/LIME)
Run Prediction



🔍 Logistic Regression — Explanations

Prediction	Probability	Confidence
No Disease	0.0173	98.3%

💬 Plain English Explanation

The Logistic Regression model predicts **no heart disease** with 98.3% confidence.

The top factors influencing this prediction are:

- **Chest Pain Type** (Typical Angina) strongly lowers the risk of heart disease.
- **Sex** (Female) moderately lowers the risk of heart disease.
- **Thalassemia Type** (Normal) moderately lowers the risk of heart disease.
- **Number of Blocked Vessels** (0 vessels) moderately lowers the risk of heart disease.
- **ST Slope** (Upsloping) slightly lowers the risk of heart disease.

✅ **Low Risk:** The model is highly confident this patient does not have heart disease.

This explanation is generated by an AI system and should be used to support, not replace, clinical judgement.

Appendix D: Decision Tree Rules

```
|--- thal <= 4.50
|   |--- ca <= 0.50
|   |   |--- oldpeak <= 3.20
|   |   |   |--- class: 0
|   |   |--- oldpeak >  3.20
|   |   |   |--- class: 1
|   |--- ca >  0.50
|   |   |--- cp <= 3.50
|   |   |   |--- class: 0
```

| | |--- cp > 3.50

| | | |--- class: 1

|--- thal > 4.50

| |--- cp <= 3.50

| | |--- chol <= 210.50

| | | |--- class: 0

| | |--- chol > 210.50

| | | |--- class: 1

| |--- cp > 3.50

| | |--- oldpeak <= 0.45

| | | |--- class: 1

| | |--- oldpeak > 0.45

| | | |--- class: 1

Appendix E: Artefact File Structure

File	Purpose
setup.py	Loads, cleans, splits, and scales the dataset
LogisticRegression.py	Trains and evaluates Logistic Regression
decision_tree.py	Trains and evaluates Decision Tree and exports rules
RandomForest.py	Trains and evaluates Random Forest
xgboost_model.py	Trains and evaluates XGBoost
shap_analysis.py	Generates SHAP explanations
lime_analysis.py	Generates LIME explanations
shap_comparison.py	Compares SHAP feature rankings across models
xai_comparison.py	Compares SHAP and LIME globally and locally
app.py	Streamlit dashboard artefact